

The missing pressure traction in Scott–Vogelius–Nitsche methods on polygonal approximations of a curved wall

Claudemir de Souza Cavalcante^a, Jaideep Sai Padhi^b

^aUniversidade Federal do Pará, Belém, PA, Brazil

^bDepartment of Computer Science, Purdue University, West Lafayette, IN 47907, USA

Abstract

We consider two-dimensional Stokes flow in which a curved no-slip wall is replaced by an inscribed polygon with edges of length at most h_Γ , and the method of Gjerde and Scott: Scott–Vogelius elements of degree $k \geq 4$ on a mesh of size h_Ω , with no-slip imposed by Nitsche’s method with penalty μ/h_Ω . Scott asked whether the H^1 error is $O(h_\Gamma^{3/2} + h_\Omega^k)$ for general Stokes data, and if not, why not. For the method as printed and a fixed penalty, the answer is no whenever the pressure varies along the wall: the Nitsche form has no pressure traction, so the discrete flow leaks through the polygon with normal velocity $(h_\Omega/\mu)(p - \bar{p})$, $\bar{p}$ the mean wall pressure. We prove that this leak is the leading error, with constant 1, and that the energy and H^1 errors are of order $(h_\Omega/\mu)^{1/2}$ and h_Ω/μ , with matching lower bounds. A mean-free traction term, whose velocity coincides with that of a zero-net-flux formulation of Frachon, Nilsson and Zahedi, attains $h_\Gamma^{3/2} + h_\Omega^k$ for general data, and the exponent 3/2 is sharp for every $k \geq 4$. A penalty scaling like $h_\Omega/\min_e |e|$ is sufficient, and necessary under a shape condition (by an exactly certified witness). Computations with $k = 4$ agree with the predicted rates, constants and threshold.

Keywords: Stokes equations, Scott–Vogelius elements, Nitsche’s method, curved boundaries, penalty parameter

1. Introduction

In 1985 Scott and Vogelius showed that for polynomial degree $k \geq 4$, on meshes without singular vertices, the divergence maps continuous piecewise- P_k velocities onto discontinuous piecewise- P_{k-1} pressures [1, 2]. The velocities are then divergence-free pointwise, with exact mass conservation and pressure-robust errors on fitted meshes [3]. At a curved wall the same property causes trouble. If the wall is replaced by an inscribed polygon and no-slip is imposed strongly, the divergence constraint can force a zero velocity gradient at a polygon vertex (for instance one lying in only two triangles), and the H^1 error is then only of order $h^{1/2}$ [4, 5] (under (M2) below this lock cannot occur [6]). Gjerde and Scott [5] therefore keep the polygon but impose the wall condition weakly, by Nitsche’s method [7], with penalty μ/h_Ω on a mesh of size h_Ω . For a zero-pressure benchmark Scott observed H^1 errors of order $h^{3/2}$, the rate expected from an uncorrected polygon (scalar corrections beyond it go back to [8]; see [9]). For it, with h_Γ the longest polygon edge, Cavalcante proved $\|u - u_h\|_{H^1(\Omega_h)} \leq C(h_\Omega^k + h_\Gamma^{3/2})$ for $k \geq 4$, under (M1)–(M2) and $\gamma_0 \leq \mu h_\Gamma/h_\Omega \leq \gamma_1$ [10, Thm. 5.2]. Scott’s prize asks for $h_\Gamma^{3/2} + h_\Omega^k$ for general Stokes data: “If this does not hold, why not?” [4].

For the method as printed it does not hold, because a term is missing. The consistent Nitsche form for Stokes flow uses the full traction, so it contains $\langle p, v \cdot n \rangle$ [11, 12]; the form printed in [5, (27)–(28)] omits it. Without a traction only the penalty

balances the wall pressure, and the discrete flow leaks through the polygon with normal velocity $u_h \cdot n \approx (h_\Omega/\mu)(p - \bar{p})$, where $\bar{p}$ is the mean of p on the wall (the mean cannot leak, as the net flux is zero). This is the consistency error of penalty methods [13], acting on the pressure alone: for fixed μ the H^1 rate is 1. Restoring the full traction fails for divergence-free spaces: the constant pressure mode then drops out of the equations and is not determined. Subtracting the wall mean of the discrete pressure repairs this. The velocity of the resulting method is that of the zero-net-flux formulation of Frachon, Nilsson and Zahedi [14, §5.1], transcribed to this setting.

Related work and novelty. Cavalcante [10] analysed the zero-pressure-gradient case, where there is no leak. His proof integrates $(\nabla u)^\top n$ by parts on each chord and uses a local Fortin operator with flux-compatible outer data; the bound also holds for every $k \geq 2$ on barycentric Clough–Tocher refinements [10, Cor. 5.3]. His quartic study finds the coercivity threshold $60.33821 < \mu^* < 60.33831$ ($\gamma^* \approx 19.07$) on a 20-chord mesh [10, (35)]. Exactly divergence-free methods of optimal order h^k exist on curved domains, at a higher cost: isoparametric Piola-mapped Scott–Vogelius elements fit the wall and are pressure-robust [15, Thm. 6.1], also in 3D [16], and Liu, Neilan and Otus [17] work inside the curved domain with a boundary multiplier, a zero-flux constraint and a Taylor correction. The placement of the pressure traction in divergence-free cut methods is studied in [18, 14]; our stability theory is that of Guzmán and Scott [2]. None of these works treats the straight-sided Gjerde–Scott method, for which $h_\Gamma^{3/2}$ is the sharp target. New here are: (i) the leak law, with constant 1 and two-sided bounds (Theorem 1); (ii) proof that the mean-free

Email addresses: claudemir.cavalcante@icen.ufpa.br (Claudemir de Souza Cavalcante), jpadhi@purdue.edu (Jaideep Sai Padhi)

traction attains $h_\Gamma^{3/2} + h_\Omega^k$ for general data (Theorem 2), and that 3/2 is sharp for every $k \geq 4$ (Remark 1); (iii) the penalty threshold $\mu \asymp h_\Omega / \min_e |e|$, sufficient, and necessary under a shape hypothesis by a certified witness (Proposition 3); (iv) $k = 4$ computations confirming all this, including the penalty study the prize requests. The failure of the naive and L_0^2 forms (Section 3) is elementary; compare [14, (5.7)–(5.9)], [17, Remark 3.2]. We treat a disk in a square; general domains, Clough–Tocher meshes, Navier–Stokes, 3D and omitted proofs are in [6]. AI use is declared at the end.

2. Setting and the three methods

Let $R = (-L, L)^2$, $L > 1$, D the unit disk, $\Omega = R \setminus \bar{D}$, $\Gamma = \partial D$, and let (u, p) be smooth (an assumption at the corners of R [6, Rem. 2.5]) with $-\Delta u + \nabla p = f$, $\operatorname{div} u = 0$ in Ω , $u|_\Gamma = 0$, $u|_{\partial R} = g$. P_h is a convex polygon with vertices on Γ , $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h \supset \Omega$. On Γ_h and on Γ , n is the unit normal pointing out of Ω_h (resp. Ω); v is the outward normal of R ; $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ product. A triangulation of Ω_h with edges $\mathcal{E}_\Gamma$ on Γ_h has $h = h_\Omega := \max \operatorname{diam} T$, $h_\Gamma := \max_{\mathcal{E}_\Gamma} |e|$, $\rho := h / \min_{\mathcal{E}_\Gamma} |e|$ and $\gamma := \mu h_\Gamma / h$, so $\mu/h = \gamma/h_\Gamma$. We assume (M0) shape regularity and $|e| \geq c_0 h_\Gamma$ on $\mathcal{E}_\Gamma$, (M1) no singular vertices, and (M2) the angle condition of [2, Thm. 1]; $h_\Gamma \ll h$ is allowed. For $k \geq 4$, V_h is the space of continuous piecewise- P_k fields, V_h^R and V_h^0 are its subspaces vanishing on ∂R and on $\partial\Omega_h$, Π_h is the space of discontinuous piecewise- P_{k-1} functions, and $Z_h := \{v \in V_h^R : \operatorname{div} v = 0\}$. Then $\operatorname{div} V_h^R = \Pi_h$, and $(V_h^0, \Pi_h \cap L_0^2)$ has an inf-sup constant β independent of h [2], [6, Lemmas 4.3, 4.5]. With $Du = \nabla u + (\nabla u)^\top$, $\partial_n u = (\nabla u)n$ and $\bar{p}_h(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q$, the Nitsche form

$$N_h(u, v) = \frac{1}{2}(Du, Dv) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + (\mu/h) \langle u, v \rangle$$

is the symmetric Nitsche form of [5]. All methods seek $u_h \in V_h$ with the outer data on ∂R and $p_h \in \Pi_h$ with $(q, \operatorname{div} u_h) = 0$ for $q \in \Pi_h$ and

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + b_\Gamma(p_h, v) = \langle \tilde{f}, v \rangle \quad \forall v \in V_h^R. \quad (1)$$

The printed method GS [5, (28)] has $b_\Gamma = 0$ (we read its test space as V_h^R and use the opposite pressure sign; see [6, §2]). The naive form has $b_\Gamma = \langle p_h, v \cdot n \rangle$, the consistent traction term. The mean-free form CNS* has $b_\Gamma = \langle p_h - \bar{p}_h(p_h), v \cdot n \rangle$. The L_0^2 variant is the naive form with $p_h, q \in \Pi_h \cap L_0^2(\Omega_h)$. The outer data are the P_k interpolant g_I of g , or $\tilde{g}_I := g_I - (m_R/8L^2)x$ with $m_R := \int_{\partial R} g_I \cdot v$, which has zero net flux, like the exact data (cf. [19]). The velocities of GS and CNS* are divergence-free pointwise.

To compare u_h with u on $\Omega_h \supset \Omega$ we extend u as a divergence-free field. Let $\tilde{u} := \operatorname{curl} \tilde{\psi}$ for a smooth extension $\tilde{\psi}$ of the stream function of u , let $\tilde{p}$ and $\tilde{f}$ extend p and f , and let $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$, which vanishes on Ω . Set $\|v\|^2 := \|\nabla v\|^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{e \in \mathcal{E}_\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2$ and $G := \|p - \bar{p}\|_{L^2(\Gamma)}$. Let $\varepsilon_h := h^k + \inf \|\tilde{u} - z\|$, the infimum over divergence-free $z \in V_h$ with the outer data. For bounded γ , $\varepsilon_h \leq Ch^k$ with data $\tilde{g}_I$, and also with g_I if $h_\Gamma \geq h^4$ (k even) or $h_\Gamma \geq h^2$ (k odd) [6,

Lemmas 4.6, 7.32]. C, c do not depend on $h, h_\Gamma, \mu, \gamma, \rho$; C_p also on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$.

The penalty must dominate the inverse trace constant on the shortest wall edge. With κ the Korn constant on V_h^R and C_I the inverse trace constant, if $\mu \geq 4\kappa C_I^2 \rho$ (true if $\gamma \geq \gamma_0 := 4\kappa C_I^2 / c_0$), then $N_h(v, v) \geq c_1 \|v\|^2$ and $|N_h(w, v)| \leq M \|w\| \|v\|$ on V_h^R , with c_1, M independent of h, μ, ρ [6, Lemma 4.2].

3. Main results

The polygon causes a consistency error $\mathcal{G}$, small as chords lie $O(h_\Gamma^2)$ from Γ and the normal tilt is odd per edge.

Lemma 1 (Consistency). *For $v \in V_h^R$, $N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = \langle \tilde{f}, v \rangle + \mathcal{G}(v)$, where $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2}) h_\Gamma^{3/2} \|v\|$ and $\mathcal{G}(v) := \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v)$.*

Proof. Use Green's formula. Chords lie within $h_\Gamma^2/4$ of Γ , so $\tilde{u} = O(h_\Gamma^2)$ on Γ_h and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$; this bounds the last three terms. On Γ , $(\nabla u)^\top n = n \operatorname{div} u = 0$. On a chord e the normal tilts by an angle $s + O(h_\Gamma^2)$, where s is the arclength from the midpoint m_e , so $(\nabla \tilde{u})^\top n = s a_e + O(h_\Gamma^2)$ with a_e constant on e . The remainder contributes $Ch_\Gamma^2 \|v\|_{L^1(\Gamma_h)}$. The leading term is $O(h_\Gamma)$ pointwise but odd in s : $|\int_e s a_e \cdot v| = |\int_e s a_e \cdot (v - v(m_e))| \leq |e|^2 \|\partial_\tau v\|_{L^1(e)}$, and an inverse trace inequality gives $Ch_\Gamma^{3/2} \|\nabla v\|$. $\square$

For $v \in Z_h$, $\int_{\Gamma_h} v \cdot n = \int_{\Omega_h} \operatorname{div} v = 0$, so for GS

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - \bar{p}, v \cdot n \rangle \quad \forall v \in Z_h. \quad (2)$$

So GS is driven by the missing load $\mathcal{R}$, and its response is, to leading order, the leak field $(h/\mu)v_\varphi$ defined next. Let $\varphi := \chi(r) \int_0^\theta (p - \bar{p}) d\theta'$ in polar coordinates, with a cut-off χ (1 near $r = 1$, 0 near ∂R); then $w := \operatorname{curl} \varphi = -(p - \bar{p})n$ on Γ . With $I_h^{C^1}$ the C^1 piecewise- P_{k+1} interpolant [20], $v_\varphi := \operatorname{curl} I_h^{C^1} \varphi \in Z_h$, $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$, and $v_\varphi = -(p - \bar{p})(x^*)n + O(h_\Gamma)$ on Γ_h , with x^* the radial projection of x onto Γ .

Theorem 1 (The printed method [6, §6]). *Let $\gamma \geq \gamma_0$ and $Y := (1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h$. (a) $\|\tilde{u} - u_h\| \leq C[(h/\mu)^{1/2} G + Y]$. If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of γ , then $\|\tilde{u} - u_h\| \geq (2M)^{-1} (h/\mu)^{1/2} G - CY$. (b) $\|\nabla(\tilde{u} - u_h)\| \leq C_p h/\mu + CY$. (c) Let $G > 0$, fix γ , and let $h \rightarrow 0$ with ρ bounded and $\varepsilon_h \leq Ch^{3/2}$. Then $\tilde{u} - u_h = (h/\mu)v_\varphi + \vartheta_h$ with $\|\vartheta_h\| = o((h/\mu)^{1/2})$ and $\|\vartheta_h\|_{\Gamma_h} = o(h/\mu)$; hence*

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1, \\ c \frac{h}{\mu} G \leq \|\nabla(\tilde{u} - u_h)\| \leq C_p \frac{h}{\mu}.$$

The H^1 rate in (b) is 1, not 1/2, because incompressibility turns the normal load into a tangential derivative, integrated by parts along the wall; an inverse estimate gives only $|\langle S, \partial_n v \rangle| \leq Ch_\Gamma^{-1/2} \|\nabla v\|$.

Lemma 2 (Normal-load cancellation). *Let τ_e, n_e be the unit tangent and normal of $e \in \mathcal{E}_\Gamma$, and S edgewise $W^{1,\infty}$ on Γ_h , continuous at vertices, with $|S \cdot \tau_e| \leq Ch_\Gamma$ on each e . Then $|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C\|\nabla v\|$ for $v \in Z_h$.*

Proof. The tangential part is at most $Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$. On e , $\text{div } v = 0$ gives $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$, so with $\sigma_e := S \cdot n_e$, $\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}$. The integral is at most $C\|v\|_{L^1(e)}$. At a vertex $x = e \cap e'$ endpoint terms combine to $v(x) \cdot (\sigma_{e'} \tau_{e'} - \sigma_e \tau_e) = O(|e| + |e'|)|v(x)|$, as S is continuous and edge directions differ by $O(|e| + |e'|)$. A 1D inverse inequality gives $\sum_e |e| \|v\|_{L^\infty(e)} \leq C|\Gamma_h|^{1/2} \|v\|_{\Gamma_h}$, and $\|v\|_{\Gamma_h} \leq C\|\nabla v\|$ uniformly in h [6, Lemma 3.2]. $\square$

Proof of Theorem 1. As the projection $\Gamma_h \rightarrow \Gamma$ has Jacobian $1 + O(h_\Gamma^2)$, $|\mathcal{R}(v)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2} |v|$. (a) For the upper bound test (2) with $z - u_h \in Z_h$, z attaining ε_h . For the lower, $\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k)$, $|v_\varphi| \leq (\mu/h)^{1/2} (G + Ch_\Gamma^2 + Ch^k) + C$, and $M[\tilde{u} - u_h] \geq (\mathcal{R}(v_\varphi) - |\mathcal{G}(v_\varphi)|)/|v_\varphi|$. (b) Write $z - u_h = \delta_R + \delta_G$ in Z_h with $N_h(\delta_R, \cdot) = \mathcal{R}$ and $N_h(\delta_G, \cdot) = \mathcal{G} - N_h(\tilde{u} - z, \cdot)$ on Z_h , so that $|\delta_G| \leq CY$. For $r_h := \delta_R - (h/\mu)v_\varphi$ and $v \in Z_h$,

$$N_h(r_h, v) = -\langle (\tilde{p} - \bar{p})n + v_\varphi, v \rangle - \frac{h}{\mu} \left[\frac{1}{2} (Dv_\varphi, Dv) - \langle \partial_n v_\varphi, v \rangle - \langle v_\varphi, \partial_n v \rangle \right].$$

On e , $(\tilde{p} - \bar{p})n + v_\varphi = (p - \bar{p})(m_e^*)\tau_e + O(h_\Gamma^2 + h^k)$, with m_e^* the projection of m_e . This is odd in s to leading order, so the first term is at most $C_p(h_\Gamma^{3/2} + h^k)\|\nabla v\|$, as in Lemma 1. Lemma 2 with $S = w$ gives $|\langle v_\varphi, \partial_n v \rangle| \leq (C_p + Ch^k h_\Gamma^{-1/2})\|\nabla v\|$. Testing with $v = r_h$ gives $|r_h| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$, and (b) follows. (c) Here $\partial_h = (\tilde{u} - z) + \delta_G + r_h$, so $|\partial_h| \leq C_p h/\mu + CY$, $\|\partial_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} |\partial_h|$, and $\|v_\varphi\|_{\Gamma_h} = G + O(h_\Gamma^2 + h^k)$. Testing (2) with v_φ , all terms but $(\mu/h)\langle \tilde{u} - u_h, v_\varphi \rangle$ and $\mathcal{R}(v_\varphi)$ are $o(G^2)$ by (a), (b), so $\|\tilde{u} - u_h\|_{\Gamma_h} \geq c(h/\mu)G$; and $\|\tilde{u} - u_h\|_{\Gamma_h} \leq C\|\nabla(\tilde{u} - u_h)\| + Ch^{k+1/2}$ by a uniform trace inequality. $\square$

Naive and L_0^2 forms. The naive system is square, and $(0, 1)$ is in the kernel of its matrix, since $-(1, \text{div } v) + \langle 1, v \cdot n \rangle = 0$ for $v \in V_h^R$. In the L_0^2 variant, $\text{div } u_h \in \Pi_h$ is orthogonal to $\Pi_h \cap L_0^2$, so $\text{div } u_h \equiv c$ with $c|\Omega_h| = \int_{\partial R} g_I \cdot v + \int_{\Gamma_h} u_h \cdot n$, and nothing forces $c = 0$: on test A below ($N = 16$), $\|\text{div } u_h\| = 4.5 \cdot 10^{-3}$. (If $c = 0$ and $\gamma \geq \gamma_2$, then u_h is the CNS* velocity, since the naive pressure form at q is the CNS* form at $q - \bar{p}_h(q)$.) The mean-free traction avoids both defects: it ignores constants, so the constant pressure mode is fixed by $(p_h, \text{div } v)$ as in GS, and the velocity stays divergence-free.

Theorem 2 (The mean-free method; [6, Thm. 7.2, Rem. 7.3]). *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$: (a) CNS* is uniquely solvable, and its velocity equals that of the naive form tested only with $\{v \in V_h^R : \int_{\Gamma_h} v \cdot n = 0\}$, as in [14]; (b) $|\tilde{u} - u_h| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$; (c) consequently, for $\gamma \leq \gamma_{\max}$ and data $\tilde{g}_I$, $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$, with C depending on $\gamma_{\max}$ (also with g_I , see Section 2).*

Sketch. Let $B^*(q, v) := -(q, \text{div } v) + \langle q - \bar{p}_h(q), v \cdot n \rangle$. Then $B^*(q + c, v) = B^*(q, v) - c \int_{\Gamma_h} v \cdot n$, and Lemma 1 gives $N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v)$ on V_h^R , where $e_u := \tilde{u} - u_h$, $e_p := p^* - p_h$ and

$p^* := \tilde{p} - \bar{p}_{\Gamma_h}(\tilde{p})$. Write $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$ (π_h the L^2 projection onto Π_h), and let $\bar{\xi}$ be the mean of ξ . Testing with V_h^0 , the inf-sup condition gives $\beta\|\xi - \bar{\xi}\| \leq C(|e_u| + h_\Gamma^{3/2})$. Testing with $z - u_h \in Z_h$ leaves $\langle \eta + \xi - \bar{\xi}, (z - u_h) \cdot n \rangle$, and $|\langle \xi - \bar{\xi}, v \cdot n \rangle| \leq C(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|v\|$ by a discrete trace inequality; this is absorbed for $\gamma \geq \gamma_2$, which proves (b). For (a), zero data give $u_h = 0$, $p_h \equiv c$, and a flux-carrying edge bubble gives $c = 0$; on zero-flux tests B^* is the naive form. (c): use a Friedrichs inequality. $\square$

Remark 1 (Further results that complete the picture, proved in [6]). (1) *Pressures and uniformity* [6, Thms. 8.11, 8.7, 7.30, 7.33]. For fixed γ , bounded ρ , the GS pressure error is $\asymp h_\Gamma^{1/2} G/\gamma$ (up to $\pm C_p h_\Gamma$). With data $\tilde{g}_I$, the CNS* pressure error is $O(h_\Gamma^{3/2} + h_\Omega^k)$ for $\gamma \leq \gamma_{\max}$, and Theorem 2(c) holds uniformly in $\gamma \geq \gamma_2$. (2) *Sharpness, every $k \geq 4$* [6, Thms. 7.8, 7.24]. Assume (S) $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$, $\theta_{\min}$ the smallest angle of the triangles touching Γ_h , and let h_Γ be small. If $W = \partial_n u \cdot \tau \not\equiv 0$ on Γ , then $\|\nabla(\tilde{u} - u_h)\| \geq c\|W\|h_\Gamma^{3/2} - Ch_\Gamma^2$ for CNS*, and for GS with any G , with c independent of γ, ρ, p . *Idea.* On $v \in Z_h$ vanishing on Γ_h with $\int_e \partial_n v = 0$, pressure and penalty drop out, and the chordal bump $-\frac{1}{2}s^2 W_e$ of $\tilde{u} \cdot \tau_e$ ($H^{1/2}$ -size $h_\Gamma^{3/2}$) pairs with the second moment of $\partial_n v \cdot \tau_e$, which an explicit C^1 piecewise- P_5 stream function on three triangles at each wall vertex makes positive when (S) holds. (3) *A growing penalty* [6, Thm. 7.30, Cor. 7.34, Thm. 7.35]. For GS, $\mu \gtrsim h_\Gamma^{-3/2}$ restores H^1 rate 3/2 (data $\tilde{g}_I$), but no scaling gives an energy rate above 1.

Scott asked how to choose the penalty. On a well-shaped triangle at a short wall edge, a divergence-free P_4 field has negative energy unless $\mu|e|/h$ is large.

Proposition 3 (Penalty threshold [6, §10]). (a) $\mu \geq 4\kappa C_I^2 \rho$ suffices (Section 2). (b) Let T have an edge $e \in \Gamma_h$ and not meet ∂R . Map it by a similarity to the triangle with base $[(0, 0), (1, 0)]$ and apex (a, b) , with barycentric coordinates λ_0 (apex) and λ_1, λ_2 . For $t \in \mathbb{R}^3$ let $v_t := \text{curl}(\lambda_1^2 \lambda_2^2 (t_0 \lambda_0 + t_1 \lambda_1 + t_2 \lambda_2)) \in P_4^*$. Let A_T, B_T, C_T be the quadratic forms $\int_T \frac{1}{2} |Dv_t|^2$, $\int_e \partial_n v_t \cdot v_t$ and $\int_e |v_t|^2$, and let λ_T be the top eigenvalue of the pencil $(2B_T - A_T, C_T)$. If $\mu|e|/h < \lambda_T$, then N_h is indefinite on Z_h for every $k \geq 4$. (c) $\lambda_T > 1$ if the apex angle is $\geq 35^\circ$ and the base angles $\geq 5^\circ$; $\lambda_T > 4, > 9.3$ for apex angles $\geq 45^\circ, \geq 60^\circ$. So if a triangle on a shortest edge of Γ_h has such a shape, coercivity on Z_h needs $\mu > \rho$.

Proof. (b) The potential vanishes to second order on the edges through the apex, so v_t , extended by zero, lies in Z_h . A_T and B_T are invariant under dilation and C_T scales with $|e|$, so $N_h(v_t, v_t) = (\mu|e|/h - \lambda_T)C_T(t) < 0$ for the top eigenvector. (c) A_T, B_T, C_T are explicit rational functions of (a, b) [6, §10.1]. On dyadic boxes covering each shape region we verify $2520 b^3 q^T (2B_T - A_T - \lambda C_T) q > 0$ for a rational q in exact interval arithmetic [6, Prop. 10.3]. For tall triangles $\lambda_T < 0$, and even one-star witnesses can fail [6, Prop. 10.6]; under (M0) alone, $\mu^* \gtrsim \rho$ is proved only with very small constants [6, Thm. 10.13]. $\square$

Table 1: H^1 seminorm errors (rate), tests A and B₁₀₀, and pressure errors $\|p^* - p_h\|_{L^2(\Omega_h)}$ for B₁₀₀. Strong: $u_h = 0$ at the nodes of Γ_h , mesh (a) (wall vertices in two or four triangles, violating (M2)). GS, CNS*: standard mesh. Rates from the preceding $N \in \{16, 24, 32, 48, 64, 96, 128, 192\}$; at $N = 256$, GS and CNS* give 1.55, 1.56 on test A.

N	test A, velocity			test B ₁₀₀ , velocity		test B ₁₀₀ , pressure	
	strong (a)	GS	CNS*	GS	CNS*	GS	CNS*
64	3.75e-1 (0.62)	1.81e-2 (1.68)	2.08e-2 (1.69)	1.88 (0.97)	6.83e-3 (1.76)	11.8 (0.65)	1.62e-2 (2.08)
96	2.98e-1 (0.59)	9.57e-3 (1.63)	1.09e-2 (1.64)	1.28 (0.98)	3.54e-3 (1.68)	9.23 (0.62)	8.01e-3 (1.80)
128	2.54e-1 (0.56)	6.11e-3 (1.59)	6.96e-3 (1.60)	9.71e-1 (0.98)	2.24e-3 (1.63)	–	–
192	–	3.27e-3 (1.57)	3.71e-3 (1.58)	6.54e-1 (0.99)	1.18e-3 (1.60)	–	–

4. Numerical study

We take $k = 4$, $L = 2.5$, and N polygon vertices radially projected from equispaced points on the square; then $h/h_\Gamma \approx 3.3$, $\max |e|/\min |e| \leq 1.97$, and each vertex of Γ_h is in three triangles. Residuals are below 10^{-12} and $\|\operatorname{div} u_h\| \leq 2 \cdot 10^{-9}$ (10^{-6} for strong, mesh (a)). We use $\mu = 100$ ($\gamma \approx 30$), known to be above the measured thresholds only (CNS*: by 27% at $N = 16$, 13% at $N = 64$, $\approx 7\%$ extrapolated [6, Rem. 15.18]). Test A is Scott’s benchmark $u = (1 - r^{-2})(-y, x)$, $p = 0$. Test B_a has stream function $(r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2y - y + x/2)$, so $G = a(7\pi/8)^{1/2}$.

In Table 1 strong imposition locks (rate $\rightarrow 1/2$), but not on our standard mesh, which satisfies (M2) (rate 1.61 at $N = 128$); on test A ($G = 0$) both Nitsche methods tend to $3/2$. On B₁₀₀ the printed method has H^1 rate 1, an error over 500 times that of CNS* at $N = 192$, and pressure rate decreasing towards $1/2$. The CNS* errors for B₁ and B₁₀₀ agree to all printed digits. On the pure-pressure test ($u = 0$, $f = \nabla p$, p as in B₁), CNS* gives $u_h = 0$ to round-off, and at $N = 96$ the ratios in Theorem 1(c) are 0.981, 0.996 ($\mu = 10^2$), 0.9993, 0.9994 ($\mu = 10^4$).

To test Proposition 3 we find μ^* by bisection on the inertia of the velocity block plus a divergence penalty. With Γ_h fixed and the box enlarged ($L = 2.5, \dots, 40$), μ^* is linear in ρ over more than a decade ($\mu^* = 59.3, \dots, 809$ for $\rho = 4.70, \dots, 63.0$ at $N = 16$; similarly at $N = 32$), so $\gamma^* \approx 18$ –21 and $\mu^*/\rho \approx 11$ –13; under refinement at $L = 2.5$, $\gamma^* = 14.7, 18.1, 19.9, 20.8$ for $N = 8, \dots, 64$. (The sweep cannot separate ρ from h/h_Γ ; collective boundary modes explain the gap to $\lambda_T > 9.3$ [6, §10.1].) Thus $\mu \geq 25\rho$ was always coercive, while a fixed μ fails once $h/h_\Gamma \gtrsim \mu/20$.

CRedit authorship contribution statement

Claudemir de Souza Cavalcante: Formal analysis (zero-pressure-gradient case, Clough–Tocher meshes), Writing – review & editing. **Jaideep Sai Padhi:** Formal analysis, Software, Validation, Writing – original draft.

Declaration of competing interest

The question studied is the subject of a prize offered by L. R. Scott; the authors intend to submit this work for it and declare no other competing interests.

Data availability

Code, raw outputs and logs for all reported numbers are at <https://github.com/jaideepsaipadhi/zero-gradient-prize> and on Zenodo, doi:10.5281/zenodo.23044296.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to develop and check proofs, write and run the numerical code, search and check the literature, and draft and edit the text. The proofs were checked by separate instances of the model acting as independent referees, the computer-assisted steps by exact-arithmetic certificates, and all numbers by reproducible public code (see Data availability); the authors reviewed these checks. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. For the results cited from [10], C. de S. Cavalcante used ChatGPT (OpenAI), as declared there.

References

- [1] L. R. Scott, M. Vogelius, Norm estimates for a maximal right inverse of the divergence operator in spaces of piecewise polynomials, *RAIRO Modél. Math. Anal. Numér.* 19 (1) (1985) 111–143.
- [2] J. Guzmán, L. R. Scott, The Scott–Vogelius finite elements revisited, *Math. Comp.* 88 (316) (2019) 515–529. doi:10.1090/mcom/3346.
- [3] V. John, A. Linke, C. Merdon, M. Neilan, L. G. Rebholz, On the divergence constraint in mixed finite element methods for incompressible flows, *SIAM Rev.* 59 (3) (2017) 492–544. doi:10.1137/15M1047696.
- [4] L. R. Scott, Relaxing gradient constraints for exactly divergence-free spaces, Prize problem statement, University of Chicago, November 13, 2023, <https://people.cs.uchicago.edu/~ridg/prizes/zerogradprize.pdf>, listed as Prize Problem Ledger problem PPL 115 (2023).

- [5] I. G. Gjerde, L. R. Scott, Verification and validation of cylinder drag: pressure and stress approximations on curved boundaries, *Adv. Comput. Sci. Eng.* 2 (3) (2024) 295–319. doi:[10.3934/acse.2024015](https://doi.org/10.3934/acse.2024015).
- [6] J. S. Padhi, Scott–Vogelius–Nitsche on a polygonally approximated boundary: convergence, the missing pressure traction, and the penalty threshold, Preprint, Zenodo, version 3 (2026). doi:[10.5281/zenodo.23044296](https://doi.org/10.5281/zenodo.23044296).
- [7] J. Nitsche, Über ein Variationsprinzip zur Lösung von Dirichlet-Problemen bei Verwendung von Teilräumen, die keinen Randbedingungen unterworfen sind, *Abh. Math. Sem. Univ. Hamburg* 36 (1) (1971) 9–15. doi:[10.1007/BF02995904](https://doi.org/10.1007/BF02995904).
- [8] J. H. Bramble, T. Dupont, V. Thomée, Projection methods for Dirichlet’s problem in approximating polygonal domains with boundary-value corrections, *Math. Comp.* 26 (120) (1972) 869–879.
- [9] T. Dupont, J. Guzmán, L. R. Scott, Obtaining higher-order Galerkin accuracy when the boundary is polygonally approximated, *J. Numer. Math.* 33 (1) (2025) 55–86. doi:[10.1515/jnma-2023-0135](https://doi.org/10.1515/jnma-2023-0135).
- [10] C. de Souza Cavalcante, Scott–Vogelius–Nitsche convergence for a zero-pressure-gradient problem on a polygonally approximated boundary, Preprint, Zenodo (2026). doi:[10.5281/zenodo.22262466](https://doi.org/10.5281/zenodo.22262466).
- [11] E. Burman, P. Hansbo, Fictitious domain methods using cut elements: III. A stabilized Nitsche method for Stokes’ problem, *ESAIM Math. Model. Numer. Anal.* 48 (3) (2014) 859–874. doi:[10.1051/m2an/2013123](https://doi.org/10.1051/m2an/2013123).
- [12] E. Burman, P. Hansbo, M. G. Larson, A cut finite element method with boundary value correction for the incompressible Stokes equations, in: F. A. Radu, K. Kumar, I. Berre, J. M. Nordbotten, I. S. Pop (Eds.), *Numerical Mathematics and Advanced Applications ENUMATH 2017*, Vol. 126 of *Lect. Notes Comput. Sci. Eng.*, Springer, Cham, 2019, pp. 183–192. doi:[10.1007/978-3-319-96415-7_15](https://doi.org/10.1007/978-3-319-96415-7_15).
- [13] I. Babuška, The finite element method with penalty, *Math. Comp.* 27 (122) (1973) 221–228. doi:[10.1090/S0025-5718-1973-0351118-5](https://doi.org/10.1090/S0025-5718-1973-0351118-5).
- [14] T. Frachon, E. Nilsson, S. Zahedi, Divergence-free cut finite element methods for Stokes flow, *BIT Numer. Math.* 64 (2024) Paper No. 39. doi:[10.1007/s10543-024-01040-x](https://doi.org/10.1007/s10543-024-01040-x).
- [15] M. Neilan, M. B. Otus, Divergence-free Scott–Vogelius elements on curved domains, *SIAM J. Numer. Anal.* 59 (2) (2021) 1090–1116. doi:[10.1137/20M1360098](https://doi.org/10.1137/20M1360098).
- [16] W. Chen, Z. Liu, Pressure-robust finite elements for the Stokes problem on three-dimensional curved domains, arXiv:2609.30008 (2026).
- [17] H. Liu, M. Neilan, M. B. Otus, A divergence-free finite element method for the Stokes problem with boundary correction, *J. Numer. Math.* 31 (2) (2023) 105–123. doi:[10.1515/jnma-2021-0125](https://doi.org/10.1515/jnma-2021-0125).
- [18] H. Liu, M. Neilan, M. Olshanskii, A CutFEM divergence-free discretization for the Stokes problem, *ESAIM Math. Model. Numer. Anal.* 57 (1) (2023) 143–165. doi:[10.1051/m2an/2022072](https://doi.org/10.1051/m2an/2022072).
- [19] F. Eickmann, L. R. Scott, T. Tscherpel, Scott–Vogelius element and iterated penalty method for inhomogeneous Dirichlet boundary conditions, arXiv:2509.17899 (2025).
- [20] J. Morgan, R. Scott, A nodal basis for C^1 piecewise polynomials of degree $n \geq 5$, *Math. Comp.* 29 (1975) 736–740.